

PAWAN SINGH

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EDUCATION

Ph.D. Geospatial Engineering

July 2021 – present

Indian Institute of Technology Roorkee, India

Supervisor: Prof. Saurabh Vijay

Dissertation: *High-resolution glacier monitoring using terrestrial data and methods*

M.Tech. Geo-Informatics (Gold Medal)

2019 – 2021

Motilal Nehru National Institute of Technology Allahabad, India

Dissertation: *Landslide detection using deep learning methods*

B.Tech. Electronics and Communication Engineering

2015 – 2019

Birla Institute of Applied Sciences, Nainital, India

RESEARCH EXPERIENCE

Research Fellow

July 2021 – present

Indian Institute of Technology Roorkee, India

Conducted multiple high-altitude field campaigns (2022–2025) in the western Himalaya, deploying terrestrial laser scanners and time-lapse camera systems for rapid glacier change monitoring, including ice-cliff dynamics and glacier ice-velocity variations. Worked on high-resolution SAR-based (testbed for the NISAR mission) glacier terrain classification. Developed strong expertise in remote sensing, terrain analysis, cryospheric field measurements, and geospatial data processing.

Project Member – Endangered Wood Architecture Preservation (EWAP)

February 2026

Uttarakhand, India

PI: Prof. Ram Sateesh Pasupuleti and Prof. Saurabh Vijay

Conducted UAV-, LiDAR-, and terrestrial photography-based surveys of two endangered wooden temple structures and developed high-resolution 3D models for documentation and preservation. Complemented geospatial data with qualitative field investigations by interviewing local priests, village heads, and residents to document the temples' historical background, cultural significance, and community value.

Project Member – Heritage Impact Assessment

June 2024 – October 2024

Punjab and Haryana High Court, Chandigarh, India

PI: Prof. Ram Sateesh Pasupuleti and Prof. Saurabh Vijay

Conducted UAV-based surveys and developed high-resolution 3D models to assess the visual and heritage impact of proposed construction projects. Performed visibility and line-of-sight analysis for UNESCO World Heritage Sites, including the Punjab and Haryana High Court, to support heritage conservation and planning decisions.

Project Member – Dumpsite Garbage Mass Estimation

October 2022 – September 2024

Department of Civil Engineering, IIT Roorkee, India

PI: Prof. Bhanu Prakash Vellanki and Prof. Saurabh Vijay

Participated in 14 large-scale dumpsite surveys (up to 0.4 km²) across Uttar Pradesh, Haryana, and Uttarakhand. Estimated garbage mass using volumetric analysis derived from drone-generated DSMs combined with in-situ density sampling, accounting for depth-wise density variation.

TEACHING EXPERIENCE

Teaching Assistant – GNSS Surveying

2020 – 2021

Motilal Nehru National Institute of Technology (MNNIT), Allahabad

Conducted tutorial sessions on MATLAB fundamentals and guided students in handling, processing, and analysis of GNSS RINEX data using MATLAB.

PUBLICATIONS

Journal Publications

1. **Singh P**, Vijay S, Banerjee A, Sarangi C, Rashid I, Zargar SA. Quantifying short-term backwasting rates of a supraglacial ice cliff at Machoi Glacier in the Indian Himalaya. *Journal of Glaciology*, 2025. doi:10.1017/jog.2024.98
2. Das G et al. Mapping glacierized regions with quad-pol dual-frequency LS-ASAR: insights for the NISAR mission. *IEEE JSTARS*. doi:10.1109/JSTARS.2025.3618572
3. **Singh P**, Vijay S, Azam M. High-frequency observation of glacier ice velocities at Drang Drung Glacier, western Himalaya. *Preprint*. SSRN: 5704778

Conference Publications

1. **Singh P**, Maurya V, Dwivedi R. Pixel-based landslide identification using Landsat 8 and Google Earth Engine. *IGARSS 2021*, Brussels, Belgium. doi:10.1109/IGARSS47720.2021.9553358
2. Saxena A, Tyagi M, **Singh P**. Digital outing system using RFID and Raspberry Pi with MQTT protocol. *IoT-SIU 2018*, Bhimtal, India. doi:10.1109/IoT-SIU.2018.8519923

FIELD EXPEDITIONS

- **Indian Himalayas:** 2022 (July, September), 2023 (June, October), 2024 (August), 2025 (June, August)
- **Landslide Surveys:** Shivalik Range (April 2021 and June 2022)
- **Architecture Survey:** Chandigarh (2024) and Uttarkashi (2026)

CONFERENCES & WORKSHOPS

International

- High-Level International Conference on Glaciers' Preservation, Dushanbe, Tajikistan, May 2025
- Debris-Covered Glacier Workshop, ISTA, Austria, November 2024
- IUGG General Assembly, Berlin, Germany, July 2023

National

- National Cryosphere and Polar Science (NCPS) Meeting, NCPOR, September 2025
- 2nd Indian Cryosphere Meet, IIT Bombay, February 2025
- 1st Indian Cryosphere Meet, IIT Ropar, February 2024
- Biodiversity and Climate Change (BDCC 2023), IIT Kharagpur, February 2023

TRAINING & SKILLS

- **Deep Learning Specialization** (DeepLearning.AI, Coursera, 2020): Neural Networks and Deep Learning; Hyperparameter Tuning and Optimization; Convolutional Neural Networks
- **Programming & Data Analysis:** MATLAB, Python, C++
- **Glacier Modelling:** Energy Balance Modelling, OGGM
- **Machine Learning:** Deep Learning, Physics-informed Deep Learning
- **Geospatial & Hardware:** QGIS, GNSS Surveying, Drone-based Surveying, LiDAR Surveying, IoT

AWARDS & HONOURS

- Early Career Travel Grant, Glacier Preservation Conference, Dushanbe 2025
- IUGG 2023 Travel Grant 2023
- AGU Cryosphere Innovation Award 2022
- Tuition Fee Waiver (TFW), Uttarakhand Technical University 2015

LEADERSHIP & COMMUNITY INVOLVEMENT

President, SPIE IITR Student Chapter *October 2022 – October 2023*

Organised online and offline expert lectures on remote sensing, optics, and related topics. Supported early-career researchers in identifying opportunities to advance their research.

Council Member, Indian Polar Research Network–APECS *October 2021 – January 2023*

Connected Indian polar researchers with early-career researchers. Organised lectures and outreach programmes to raise awareness of the climate emergency.

Volunteer, IGARSS 2021 IEEE *Brussels, Belgium · April 2021*

Served as session co-chair for the session on landslide detection using modern methods.

Volunteer, An Hour of Code – Microsoft *Nainital · December 2017*

Organised a workshop introducing primary school students to coding using interactive tools.

MEMBERSHIPS

- International Glaciological Society (IGS)
- International Association of Cryospheric Sciences (IACS)
- American Geophysical Union (AGU)
- European Geophysical Union (EGU)

OTHER ACHIEVEMENTS & INTERESTS

- Actively engaged in long-distance running, swimming, hiking, and rowing.
 - National medalist in Dragon Boat Racing.
 - North-zone winner in IoT competition organised by IIT Bombay, Varanasi.
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